

Mohammed Khalid Shaik

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Software engineer with 6 years of experience building APIs and data-intensive applications. Currently engineering security, observability, and CI/CD pipelines for commercial agentic AI platform used in life sciences research. Previously owned large-scale data infrastructure for companies including Takeda and Crinetics along with leading their AWS disaster recovery under formal change control and GxP audit.

SKILLS

AI / LLM Engineering: Agent orchestration (LangGraph, Pydantic AI) · Model routing (LiteLLM) · Semantic caching (LangCache) · Retrieval (RAG, pgvector) · Sandboxed agent execution (Docker Sandboxes, macOS Seatbelt, Windows AppContainer) · **Frameworks:** Tauri, FastAPI, React · **Languages:** Python, Rust, JavaScript/TypeScript, SQL · **Backend & APIs:** REST, WebSockets, async & concurrent programming, microservices, distributed systems · **Cloud & Infra:** AWS, Terraform, Kubernetes, Docker · **Data:** PostgreSQL, Redis, Pandas, Pydantic, SQLAlchemy, Airflow · **Testing & Quality:** pytest, Locust load testing, integration & e2e testing, AST static analysis, CI/CD · **Compliance:** GxP, change control, audit readiness

PROFESSIONAL EXPERIENCE

Software Engineer, Integrated Analytics Solutions

May 2023 – Present

Sentinel, AI agent platform for life sciences research

- Built least-privilege sandboxing for coding agents on Sentinel, a life sciences AI agent platform used by 30+ drug researchers. It uses macOS Seatbelt, chosen after ruling out Apple's App Sandbox, so each agent can only reach the files its task allows
- Built a 10-trial evaluation that blocks new agent personalities whose behavior breaks their guardrails, and added Codex and Hermes to Sentinel's agent monitoring, covering token spend and blocked sessions.
- Designed a multi-agent system for shipping production fixes and security patches, controlling what context each agent receives, isolating workers in Docker Sandboxes, tracking each agent's token spend, and gating every merge on a read-only verifier agent's test proof, cutting reopened tickets by about 50%.

Crinetics Pharmaceuticals, clinical-data delivery

- Rebuilt Crinetics' regulated clinical-data delivery (SFTP to SMB) as an agent-operable service that AI agents and administrators can configure, monitor, diagnose and recover. It handles thousands of files a day and delivers ~100-file bursts in under 3 seconds with exactly-once delivery and immutable GxP audit trails

Takeda, R&D Data infrastructure

- Owned AWS disaster recovery for a single-region R&D SaaS platform (1,000+ users, ~20 TB), delivering a GxP-validated 2-hour RTO / 4-hour RPO. Authored the standby environment in Terraform and Lambda automation that auto-attached backup EBS volumes to recovery instances at failover.
- Served as the approving reviewer on the data infrastructure Terraform repository, where every applied plan mutated AWS Infra. Gated 50+ PRs with validated testing, documented releases, and change-control sign-off, enabling junior engineers to ship safely in a GxP environment.
- Built a FastAPI and SAS automation service that cut manual FDA clinical report RTF-to-PDF assembly from 15+ minutes to under 30 seconds per (~10 MB) file, supporting RTFs up to 1 GB for 10,000+ statistical users.

Software Engineer, GameStop

Nov 2022 – May 2023

- Automated promotion approvals with Python, SQL, and REST APIs, cutting monthly review effort by 80% (100 hours (about 8 days) → 20 hours) and eliminating the manual gaps that had previously exposed the team to compliance penalties.

Software Engineer, M&G

Jan 2021 – Aug 2022

- Developed a service for monitoring and self-healing across 1,500+ servers, resolving about half of recurring tickets automatically and saving 100+ team hours per month while protecting SLA timelines.

EDUCATION

- Masters in data science, The University of Texas at Arlington, Dec 2023